

HKCEE 1983
Mathematics II

Label 1 $\frac{6}{x^2-9} - \frac{5}{x^2+x-6}$

- A. $\frac{1}{(x-2)(x-3)}$
 B. $\frac{1}{(x+2)(x+3)}$
 C. $\frac{1}{(x+2)(x-3)}$
 D. $\frac{1}{(x-2)(x+3)}$
 E. $\frac{x-27}{(x-2)(x+3)(x-3)}$

Label 2 $\frac{1}{\frac{1}{a^3} + \frac{1}{b^3}}$
 Figure 2a Density 356

- A. $\frac{1}{a^2} + \frac{1}{b^2}$
 B. $\frac{1}{a^2} + \frac{1}{ab} + \frac{1}{b^2}$
 C. $\frac{1}{a^2} - \frac{1}{ab} + \frac{1}{b^2}$
 D. $\frac{a^2 - ab + b^2}{a^2 + ab + b^2}$
 E. $\frac{a^2 + ab + b^2}{a^2 - ab + b^2}$

Figure 2b Density 172

Label 3 If $x = \frac{y^2}{\sqrt{a^2+bz}}$, then $z =$

- A. $\frac{1}{b} \left(\frac{y^4}{x^2} - a^2 \right)$
 B. $\frac{1}{b} \left(\frac{x^2}{y^4} - a^2 \right)$
 C. $\frac{1}{b} \left(a^2 - \frac{x^2}{y^4} \right)$
 D. $\frac{1}{b} \left(a^2 - \frac{y^4}{x^2} \right)$

E. $\frac{1}{b} \left(a^2 - \frac{x^2}{y^2} \right)$

Label 4 $(x^2y^{-1}) \div (x^{\frac{1}{2}}y^{-1})^2 =$

- A. xy
 B. xy^{-1}
 C. xy^{-3}
 D. x^2y^2
 E. $x^{\frac{1}{2}}y^{-2}$

Label 5 The H.C.F. of a^3-1 and a^4-1 is

- A. 1
 B. $a \pm 1$
 C. $a-1$
 D. a^2+1
 E. a^2-1

Label 6 When $f(x)$ is divided by $(2x+1)$, the remainder is

- A. $f(2)$
 B. $f(1)$
 C. $f(-1)$
 D. $f\left(\frac{1}{2}\right)$
 E. $f\left(-\frac{1}{2}\right)$

Label 7 If α and β are the roots of $2x^2-3x-4=0$, then $\alpha^2+3\alpha\beta+\beta^2$

- A. 1
 B. 4
 C. $\frac{1}{4}$
 D. $\frac{1}{5}$

D. $8\frac{1}{4}$
E. 13

Label 8 $2x - 3a - 4 > 3x + 5a + 6$ is equivalent to

- A. $x > -8a - 10$
 B. $x > 2a - 10$
 C. $x \leq -8a - 10$
 D. $x < \frac{1}{5}(2a + 2)$
 E. $x > \frac{1}{5}(2a + 2)$

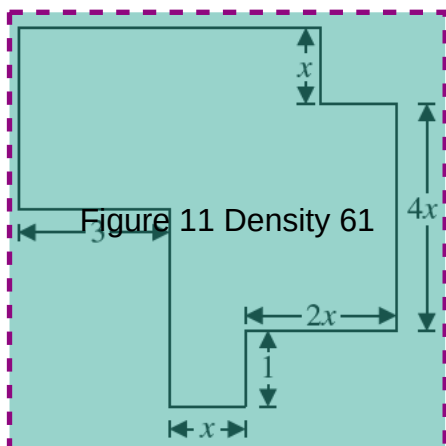
Label 9 The sixth term and the eleventh term of an A.P. are 10 and 30 respectively. The first term is

- | | |
|----|-----|
| A. | -14 |
| B. | -10 |
| C. | 10 |
| D. | 50 |
| E. | 54 |

Label 10 If $2x = 3y = 5z$, then $x : y : z =$

- A. $2:3:5$
B. $5:3:2$
C. $6:10:15$
D. $15:10:6$
E. $25:9:4$

Label 11



In the figure, all the corners are right-angled. If the perimeter of the figure is 40, then $x =$

- | | |
|----|------|
| A. | 0.25 |
| B. | 2 |
| C. | 2.5 |
| D. | 4 |
| E. | 4.5 |

Label 12 If the lengths of the diagonals of a rhombus are 2 cm and 4 cm respectively, what is the area of the rhombus?

- A. 2 cm^2
B. 4 cm^2
C. 8 cm^2
D. 16 cm^2
E. It cannot be determined

Label 13 A hollow cylindrical metal pipe, 1 m long, has an external radius and an internal radius of 5 cm and 4 cm respectively. The volume of metal is

- A. $90\pi \text{ cm}^3$
B. $100\pi \text{ cm}^3$
C. $180\pi \text{ cm}^3$
D. $900\pi \text{ cm}^3$
E. $1800\pi \text{ cm}^3$

Label 14 Two men cycle round a circular track which 3 km long. If they start at the same time and at the same spot but go in opposite direction with speeds 6km/h and 9 km/h respectively, for how long must they cycle before they meet for the first time?

- A. 12 minutes
B. 15 minutes
C. 18 minutes
D. 24 minutes
E. 60 minutes

Label 15
15. A man marks his good at a price that will bring him a profit of 25% on the cost price. If he wants to sell his goods to a friend at the cost price, the percentage discount on the marked price should be

- A. 25% .
- B. 20% .
- C. $16\frac{2}{3}\%$
- D. 15% .
- E. 12% .

Label 16
16. $\sin^2 \theta - (\sin^2 \theta \cos^4 \theta + \sin^4 \theta \cos^2 \theta) =$

- A. $\sin^4 \theta$
- B. $\cos^4 \theta$
- C. $-\sin^4 \theta$
- D. $-\cos^4 \theta$
- E. $\sin^2 \theta \cos^2 \theta$

Label 17
17. $\frac{\cos(90^\circ - \theta)}{\tan(180^\circ - \theta)} =$

- A. $\cos \theta$
- B. $-\cos \theta$
- C. $-\frac{\sin^2 \theta}{\cos \theta}$
- D. $-\frac{\cos^2 \theta}{\sin \theta}$
- E. $\frac{\sin^2 \theta}{\cos \theta}$

Label 18
18.

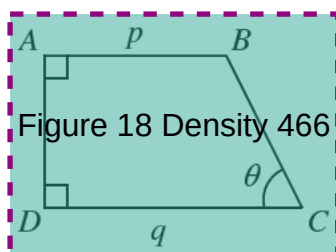


Figure 18 Density 466

In the figure, $AB = p$, $DC = q$ and $\angle A = \angle D = 90^\circ$. $BC =$

- A. $(q - p) \sin \theta$

- B. $(q - p) \cos \theta$
- C. $(q - p) \tan \theta$
- D. $\frac{q - p}{\sin \theta}$
- E. $\frac{q - p}{\cos \theta}$

Label 19
19.

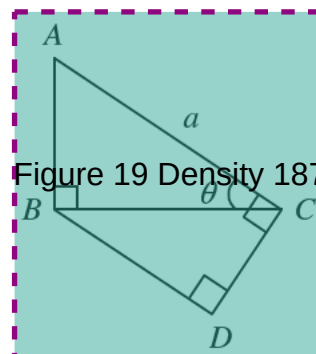


Figure 19 Density 187

In the figure, $\angle ABC = \angle ACD = \angle BDC = 90^\circ$. $AC = a$. $CD =$

- A. $a \sin^2 \theta$
- B. $a \sin^2 \theta$
- C. $a \tan \theta$
- D. $a \sin \theta \cos \theta$
- E. $\frac{a \cos \theta}{\sin \theta}$

Label 20
20.

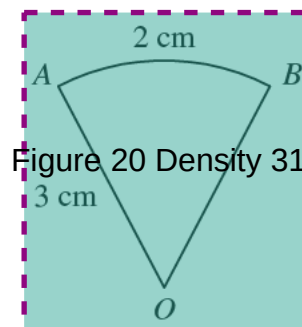


Figure 20 Density 312

In the figure, OAB is a sector of a circle. Radius OA is 3 cm long and arc $AB = 2$ cm. The area of the sector is

- A. 3 cm^2
- B. 6 cm^2
- C. 9 cm^2
- D. $3\pi \text{ cm}^2$
- E. $6\pi \text{ cm}^2$

Label 21

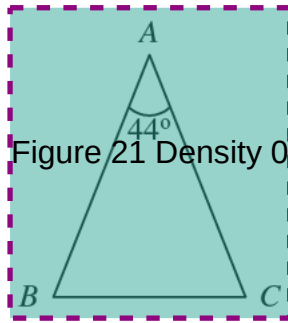


Figure 21 Density 0

The sum of the six marked angles in the figure is

- A. 360°
- B. 540°
- C. 600°
- D. 720°
- E. 900°

In the figure, $AB = AC$. If the area of $\triangle ABC$ is 64 cm^2 , then $AB =$

- A. 32 cm
- B. $16\sqrt{2}$ cm
- C. 16 cm
- D. $8\sqrt{2}$ cm
- E. 4 cm

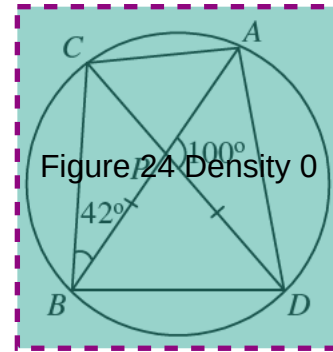


Figure 24 Density 0

Label 22

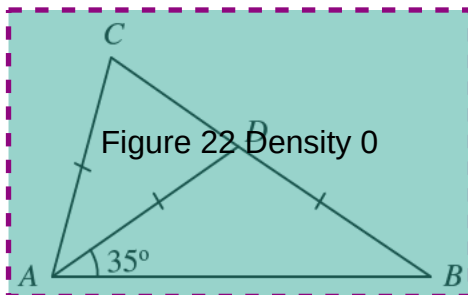


Figure 22 Density 0

In the figure, chords AB and CD intersect at P . $BP = DP$. $\angle CAD =$

- A. 58°
- B. 86°
- C. 88°
- D. 92°
- E. 142°

In the figure, D is a point on BC and $AC = AD = BD$. $\angle CAD =$

- A. 20°
- B. 25°
- C. 30°
- D. 35°
- E. 40°

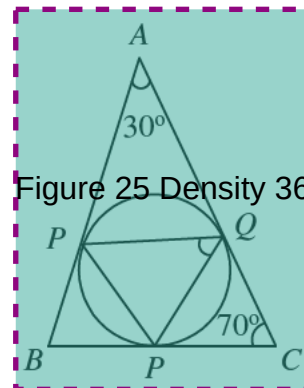


Figure 25 Density 36

Label 23

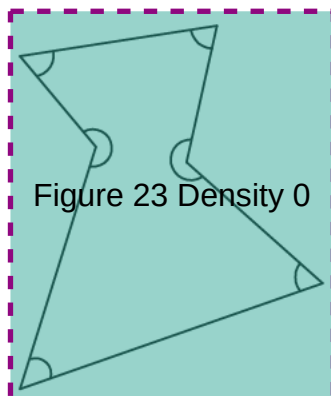


Figure 23 Density 0

In the figure, the three sides of $\triangle ABC$ touch the circle at the points P , Q and R . $\angle PQR =$

- A. 30°
- B. 50°
- C. 55°
- D. 70°
- E. 75°

Label 26
26. If the line $2x - 3y + c = 0$ passes through the point $(1, 1)$, then $c =$

- A. -2
- B. -1
- C. 0
- D. 1
- E. 2

Label 27
27. The equation of the line passing through $(1, -1)$ and perpendicular to the x -axis is

- A. $x - 1 = 0$
- B. $x + 1 = 0$
- C. $y - 1 = 0$
- D. $y + 1 = 0$
- E. $x + y = 0$

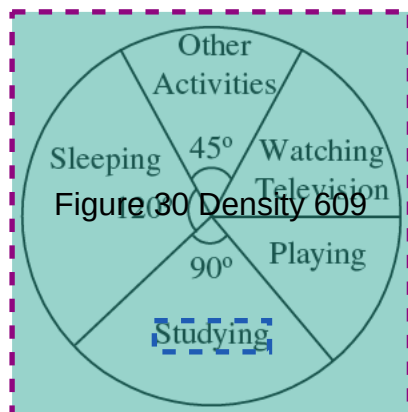
Label 28
28. A circle has its centre at $(3, 4)$ and passes through the origin. Its equation is

- A. $x^2 + y^2 - 6x - 8y + 25 = 0$
- B. $x^2 + y^2 - 3x - 4y = 0$
- C. $x^2 + y^2 - 6x - 8y = 0$
- D. $x^2 + y^2 + 6x + 8y = 0$
- E. $x^2 + y^2 - 6x - 8y + 25 = 0$

Label 29
29. If d is the distance between the point (a, b) and (b, a) , then d^2

- A. 0
- B. $a^2 + b^2$
- C. $2(a^2 + b^2)$
- D. $(a - b)^2$
- E. $2(a - b)^2$

Label 30
30.



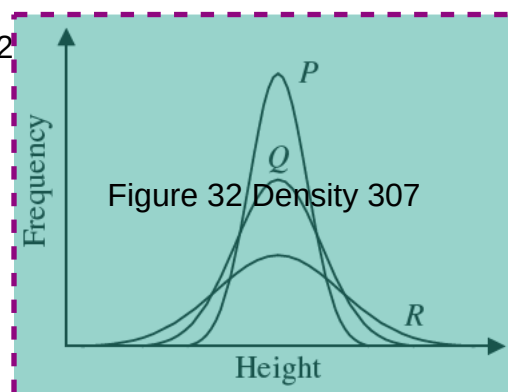
The pie chart shows how a boy spends the 24 hours of a day. If the boy spends 4 hours playing, how much time does he spend watching television?

- A. 1 hour
- B. 2 hours
- C. 3 hours
- D. 4 hours
- E. 5 hours

Label 31
31. There are 12 boys and 8 girls in a class, $\frac{1}{4}$ of the boys and $\frac{1}{4}$ of the girls wear glasses. What is the probability that a student chosen at random from the class is a boy not wearing glasses or a girl wearing glasses?

- A. $\frac{5}{20}$
- B. $\frac{9}{20}$
- C. $\frac{11}{20}$
- D. $\frac{15}{20}$
- E. $\frac{9}{100}$

Label 32
32.



In the figure, P , Q and R are curves showing the frequency distributions of the heights of students in three schools, each having the same number of students. Which distribution has the greatest standard deviation and which the smallest?

	Greatest	Smallest
A.	P	Q
B.	P	R
C.	Q	R
D.	R	P
E.	R	Q

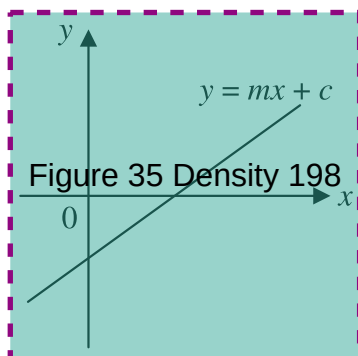
Label 33
33. If $x + \frac{1}{x} = 2 + \frac{1}{2}$, then $x =$

- A. 2 only
- B. -2 only
- C. $\frac{1}{2}$ only
- D. -2 or 2
- E. $\frac{1}{2}$ or 2

Label 34
34. $2 - x - x^2 < 0$ is equivalent to

- A. $x < -4$
- B. $x > 3$
- C. $-4 < x < 3$
- D. $x < -3$ or $x > 4$
- E. $x < -4$ or $x > 3$

Label 35
35.



In the figure, the equation of the straight line is $y = mx + c$. Which one of the following is true?

- A. $m > 0$ and $c > 0$
- B. $m > 0$ and $c < 0$
- C. $m < 0$ and $c > 0$
- D. $m < 0$ and $c < 0$
- E. $m > 0$ and $c = 0$

Label 36
36. If a and b are positive numbers, which of the following is/are true?

- I. $\log_{10}(a + b) = \log_{10}a + \log_{10}b$
- II. $\log_{10} \frac{a}{b} = \log_{10}a - \log_{10}b$
- III. $\frac{\log_{10} a}{\log_{10} b} = \frac{a}{b}$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. I, II and III

Label 37
37. A function $f(x)$ is called an even function if $f(x) = f(-x)$. Which of the following function is/are even functions?

- I. $f_1(x) = \frac{1}{x}$
- II. $f_2(x) = x^2$
- III. $f_3(x) = x^3$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

Label 38
38. In an arithmetic progression, the first term is 3 and the common difference is 2. If the sum of the first n terms of the arithmetic progression is 143 then $n =$

- A. 10
- B. 11
- C. 12
- D. 13
- E. 14

Label 39 Three positive numbers a , b and c are in geometric progression. Which of the following are true?

- I. $\frac{1}{a}$, $\frac{1}{b}$, $\frac{1}{c}$ are in geometric progression.
 - II. a^2 , b^2 , c^2 are in geometric progression.
 - III. $\log_{10}a$, $\log_{10}b$, $\log_{10}c$ are in arithmetic progression.
- A. I and II only
 - B. I and III only
 - C. II and III only
 - D. I, II and III
 - E. None of them

Label 40 The scale of a map is 1 : 20 000. On the map the area of a farm is 2 cm^2 . The actual area of the farm is

- A. 400 m^2
- B. 800 m^2
- C. $40\,000 \text{ m}^2$
- D. $80\,000 \text{ m}^2$
- E. $8\,000\,000 \text{ m}^2$

Label 41 It took Paul 40 minutes to walk from Town A to Town B. If the return journey took him 30 minutes, the percentage increase in his speed was

- A. 10%.
- B. $16\frac{2}{3}\%$
- C. 25%.
- D. $33\frac{1}{3}\%$
- E. 40%.

Label 42 A merchant sold 2 articles each at \$1000. For first article, he gained 25% on the cost price. For the second article, he lost 20% on the cost price. Altogether

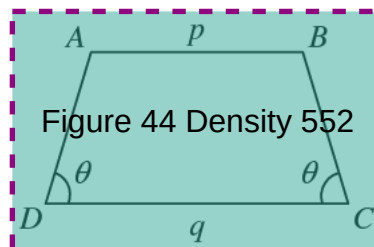
- A. he gained \$100.

- B. he gained \$50
- C. he lost \$100.
- D. he lost \$50.
- E. he lost \$48.

Label 43 Three number are in the ratio 2 : 3 : 5. The ratio of their average to the largest of the three numbers is

- A. 1 : 3
- B. 1 : 2
- C. 3 : 5
- D. 2 : 3
- E. 2 : 1

Label 44



In the figure, $ABCD$ is a trapezium in which $AB \parallel DC$ and $\angle C = \angle D = \theta$. If $CD = p$ and $AB = q$, then the area of the trapezium is

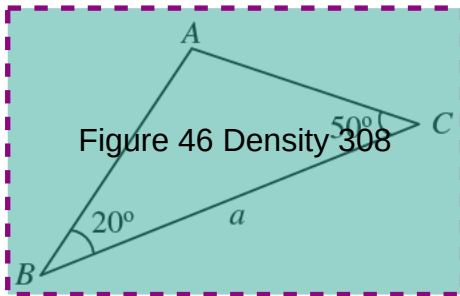
- A. $\frac{1}{2}(p+q)^2 \tan \theta$.
- B. $\frac{1}{4}(p^2+q^2) \tan \theta$.
- C. $\frac{1}{2}(p^2-q^2) \tan \theta$.
- D. $\frac{1}{4}(p^2-q^2) \tan \theta$.
- E. $\frac{(p^2-q^2)}{4 \tan \theta}$.

Label 45 A solid sphere is cut into two hemispheres. The percentage increase in the total surface area is

- A. 25%.
- B. $33\frac{1}{3}\%$
- C. 50%.
- D. 75%.

E. 100%

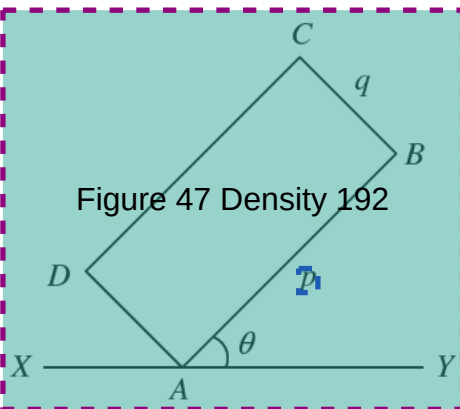
Label 46
46



In the figure, $BC = a$. $AB =$

- A. $a \sin 20^\circ$
- B. $\frac{a \sin 20^\circ}{\sin 70^\circ}$
- C. $\frac{a \sin 20^\circ}{\sin 50^\circ}$
- D. $\frac{a \sin 50^\circ}{\sin 20^\circ}$
- E. $\frac{a \sin 50^\circ}{\sin 70^\circ}$

Label 47
47



In the figure, $ABCD$ is a rectangle. $AB = p$ and $BC = q$. If $\angle BAY = \theta$, the distance of C from the XY is

- A. $(p + q) \sin \theta$.
- B. $(p + q) \cos \theta$.
- C. $\sqrt{p^2 + q^2} \sin \theta$
- D. $p \cos \theta + q \sin \theta$
- E. $p \sin \theta + q \cos \theta$.

Label 48
48

If $0 \leq \theta < 360^\circ$, the number of roots of the equation $4 \sin^2 \theta \cos \theta = \cos \theta$ is

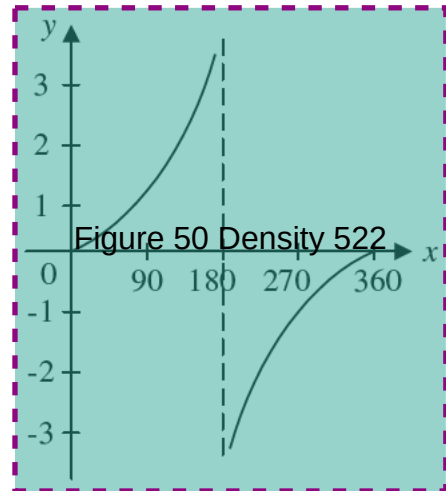
- A. 2
- B. 3
- C. 4
- D. 5
- E. 6

Label 49
49

The maximum value of $\cos^2 3x$ is

- A. 1
- B. 2
- C. 3
- D. 6
- E. 9

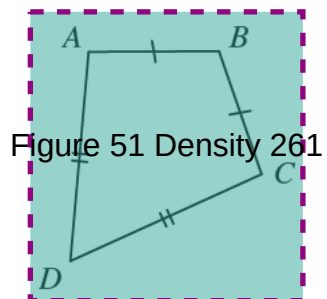
Label 50
50



The figure above shows the graph of a tangent function from 0° to 360° . The function is

- A. $y = \tan \frac{x^\circ}{2}$
- B. $y = \tan x^\circ$
- C. $y = \tan 2x^\circ$
- D. $y = \tan(x - 90)^\circ$
- E. $y = \tan(x + 90)^\circ$

Label 51
51

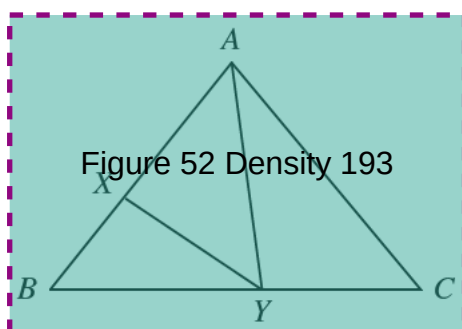


In the figure, $ABCD$ is a quadrilateral, with $AB = BC$ and $AD = DC$. Which of the following is/are true?

- I. $\angle BAD = \angle BCD$
- II. $AC \perp BD$
- III. BD bisect AC

- A. I only
- B. I and II only
- C. I and III only
- D. II and III only
- E. I, II and III

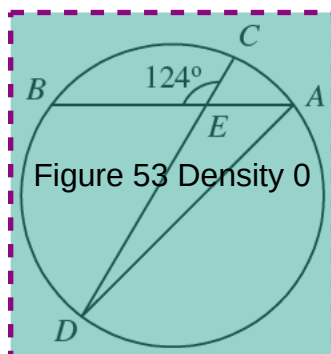
Label 52



In the figure, X and Y are points on AB and BC respectively such that $AX : XB = 3 : 2$ and $BY : YC = 4 : 3$. If the area of $\triangle ABC = 70$, then the area of $\triangle AXY =$

- A. 16
- B. 24
- C. 30
- D. 40
- E. 42

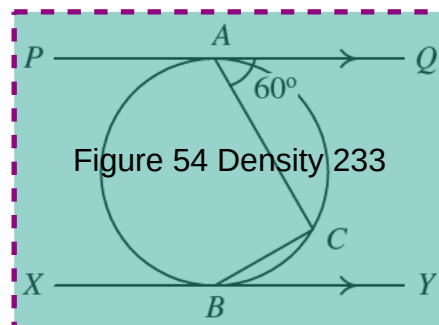
Label 53



In the figure, chords AB and CD intersect at E . The length of the minor arc BD is three times the length of the minor arc AC . $\angle BAD =$

- A. 31°
- B. 35°
- C. 42°
- D. 45°
- E. 56°

Label 54



In the figure, PQ and XY touch the circle at A and B respectively. $PQ \parallel XY$ and $\angle QAC = 60^\circ$. $\angle CBX =$

- A. 150°
- B. 135°
- C. 120°
- D. 110°
- E. 100°